

PROJECT MANUAL

AI-Enhanced Intrusion Detection System

Student Name: Sourabh Bhimrao Patil

Project Title: AI-Enhanced Intrusion Detection System

Project Description

The AI-Enhanced Intrusion Detection System (IDS) is a machine learning-based cybersecurity application developed to detect and classify network intrusions in real time. The system utilizes a Random Forest Classifier trained on NSL-KDD dataset features to identify and categorize network traffic into five classes: Normal, DoS, Probe, R2L, and U2R attacks.

The application is built using Python Flask for backend processing, Bootstrap 5 for the user interface, and Scikit-learn for machine learning implementation. The system provides a dashboard for monitoring activities, a prediction module for analyzing network traffic, an alert logging mechanism, and a REST API for external integration.

Milestone 1: Project Initiation and Planning

Defining Scope and Objectives

The primary objective of this project is to design and implement an intelligent intrusion detection system capable of accurately identifying malicious network activities. The project focuses on improving cybersecurity through machine learning-driven threat detection.

Stakeholder Identification and Engagement

Stakeholders include:

- Network Administrators
- Security Analysts
- System Administrators
- Academic Evaluators
- End Users

Project Plan and Timeline

The project was divided into phases: 1. Requirement Analysis 2. Data Collection 3. Feature Engineering 4. Model Development 5. Model Evaluation 6. Web Application Development 7. Testing and Validation 8. Deployment

Resource Allocation and Budgeting

Resources Used:

- Python Programming Language
 - Flask Framework
 - Scikit-learn Library
 - Bootstrap Framework
 - NSL-KDD Dataset
 - Personal Computer for Development
-

Milestone 2: Data Collection and Preparation

Identifying Relevant Data Sources

The NSL-KDD dataset was selected as the primary source of network traffic data for training and testing the intrusion detection model.

Data Gathering and Acquisition

Dataset records containing normal and attack traffic patterns were collected and organized for processing.

Data Preprocessing and Cleaning

The dataset was cleaned by:

- Removing inconsistencies
- Handling missing values
- Eliminating irrelevant records

Data Normalization and Transformation

Feature values were normalized using StandardScaler to improve machine learning performance and consistency.

Milestone 3: Feature Engineering and Selection

Feature Extraction Techniques

Ten important features were selected:

1. Duration
2. Source Bytes
3. Destination Bytes
4. Land
5. Wrong Fragment
6. Urgent
7. Hot
8. Number of Failed Logins
9. Logged In
10. Number of Compromised Conditions

Dimensionality Reduction

Only the most relevant intrusion-related features were retained to improve efficiency and reduce complexity.

Feature Relevance and Significance

Selected features directly influence attack classification and contribute significantly to intrusion detection accuracy.

Milestone 4: Model Selection and Training

Choosing Machine Learning Algorithms

Various algorithms were considered, including:

- Decision Trees
- Support Vector Machines
- Neural Networks
- Random Forest

Random Forest was selected due to its high accuracy and robustness.

Data Splitting and Preparation

Dataset Split:

- 80% Training Data

- 20% Testing Data

Algorithm Implementation and Configuration

The Random Forest Classifier was configured with 100 decision trees.

Model Training and Optimization

The model was trained using processed NSL-KDD data and optimized through parameter tuning.

Milestone 5: Model Evaluation and Optimization

Performance Metrics Selection

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1 Score

Model Evaluation on Validation Set

The trained model was evaluated using test datasets to measure prediction performance.

Hyperparameter Tuning and Cross-Validation

Model parameters were adjusted to maximize detection accuracy while minimizing false positives.

Model Comparison and Selection

Random Forest achieved the best performance and was selected as the final model.

Milestone 6: Real-Time Monitoring and Analysis

Real-Time Data Ingestion

The application accepts network feature values through a web interface and API requests.

Pattern Recognition and Anomaly Detection

Incoming traffic data is analyzed and classified using the trained Random Forest model.

Real-Time Alerts and Notifications

Suspicious traffic generates alerts and is logged for monitoring and investigation.

Milestone 7: Automated Classification and Alerting

Intrusion Classification Criteria

The system classifies traffic into:

- Normal Traffic
- DoS Attack
- Probe Attack
- R2L Attack
- U2R Attack

Algorithmic Classification Implementation

Prediction results are generated using machine learning inference after preprocessing.

Automated Alert Generation

Detected attacks are automatically recorded with:

- Timestamp
 - Attack Type
 - Confidence Score
-

Milestone 8: Continuous Learning and Maintenance

Integration of Threat Intelligence

Future enhancements can incorporate updated threat intelligence feeds.

Regular Model Updates and Retraining

The model can be retrained periodically using updated intrusion datasets.

Performance Monitoring and Feedback Loop

Prediction logs and system feedback can be analyzed to improve performance.

Milestone 9: Testing and Validation

Scenario-Based Testing

The system was tested using simulated intrusion scenarios.

Test Data Generation

Different combinations of feature values were used to validate predictions.

Performance Metrics Assessment

The system achieved approximately 96–100% accuracy on the testing dataset.

Validation Against Real-World Data

The design supports integration with real network traffic for future validation.

Milestone 10: Deployment and Integration

Production Environment Preparation

Deployment requirements:

- Python 3.x
- Flask Framework
- Scikit-learn
- Required Dependencies

System Integration with Existing Infrastructure

The application can be integrated with network monitoring systems and security tools.

Integration Testing

All modules including dashboard, prediction system, alerts, and API were tested successfully.

Scalability and Resource Allocation

The modular architecture supports future scaling and enhancements.

Milestone 11: Training and Knowledge Transfer

Training Materials Development

User documentation and setup instructions were prepared.

Training Sessions and Workshops

Users can be trained to:

- Monitor alerts
- Analyze predictions
- Use API services

Hands-On Demonstrations

Demonstrations were conducted using sample intrusion data and attack simulations.

Milestone 12: Ongoing Monitoring and Improvement

Continuous Monitoring and Performance Assessment

System performance can be monitored through prediction logs and dashboard statistics.

Feedback Collection and Analysis

User feedback can be collected to improve detection accuracy and usability.

Model Updating and Re-Training Strategy

Future datasets can be incorporated to improve attack detection capabilities.

Maintenance and Support Procedures

Regular maintenance includes:

- Updating dependencies
 - Retraining models
 - Monitoring logs
 - Fixing security vulnerabilities
-

Conclusion

The AI-Enhanced Intrusion Detection System successfully demonstrates the practical application of machine learning in cybersecurity. Using a Random Forest Classifier and NSL-KDD dataset, the system accurately detects and classifies network intrusions into multiple attack categories. The integration of a Flask web application, alert management, dashboard visualization, and REST API provides a complete solution suitable for academic projects, internship demonstrations, and cybersecurity research.